OpenMycoNet – COMBO_NODE

REVISION: 3.6
VARIANT: *NODE_VARIATIONS*
STATUS: *READY_FOR_MANUFACTURING*

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RELEASED STATUS PROJECT

DONE

DATE: 2026 – 10 – 25

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PREVIOUS TARGET CHECKLIST (3.5)

[■] POINT 06: 1.65V MIDPOINT REFERENCE DISTRIBUTION (VMID_REFF)
- Connect R44 (10k) and C34 (100n) bottom pins to GND.
- Route 1.65V node via R62 (22 Ω isolation resistor) to net VMID_REFF.
- Distribute VMID_REFF to:
  1) INA333 (U8) Pin 5 (REF)
  2) Input bias network R32
  3) OPA2320 (U13B) Pin 7 (+INB)
  4) Stimulation return (J11 Pin 2 & bottom of R46)

[■] POINT 07: EC COUPLING CAPACITOR UPGRADE (C28)
- Change C28 from 1 µF to 100 µF (1206 MLCC, >= 6.3V rating).
- Lowers 16 Hz AC capacitive reactance (Xc) from ~9.95 kΩ to ~99.5 Ω.

[■] POINT 08: ONE-WIRE TEMP SENSOR PULL-UP (J8 / DS18B20)
- Add 4.7 kΩ pull-up resistor between DS18B20_DATA and +3.3V rail.

[■] POINT 09: C22 GROUND NET CORRECTION
- Re-route C22 ground pin from digital GNDA to intended GND net.

[■] POINT 10: LED POWER SUPPLY RE-INTEGRATION
- Reconnect LED current-limiting resistors to +BATT supply (not VDD).

[■] POINT 11: INA333 BIAS-RETURNS, U13 DECOUPLING & SHEET NOTES
- Add 10 MΩ bias-return resistors (R63, R64) from INA+ and INA- to VMID_REFF.
- Add 100 nF decoupling capacitor (C35) between U13 Pin 10 (+3.3VA) and GNDA.
- Update DIGI_&_ANA_SENSORS design text notes to reflect current Rev 3.5 schematic.

FINAL REVISION TARGET CHECKLIST (3.6)

[■] POINT 12: TH1 THERMISTOR DNP ASSIGNMENT
- Set TH1 attribute to DNP (Do Not Populate) in the schematic.
- Verify TH1 is completely excluded from the final BOM and CPL files.

[■] POINT 13: ANALOG RAIL ISOLATION (FB1 REMOVAL)
- Remove FB1 bridging +3.3V and +3.3VA between U2 and U3 outputs.
- Verify regulator outputs are physically and electrically separated to guarantee clean analog rail integrity.

[■] POINT 14: VMID_REFF BUFFER & QUIESCENT CURRENT REDUCTION
- Insert a low-quiescent-current voltage follower (buffer) between the C34 midpoint and the VMID_REFF distribution node.
- Increase voltage divider resistor values (R43/R44) to reduce continuous quiescent current consumption.

[■] POINT 15: DESIGNATOR ALIGNMENT, LAYOUT & BOM CLEANUP
- PCB Layout: Ensure the 100 nF decoupling cap (C48) is physically placed as close to U13 Pin 10 as practical.
- Documentation Sync: Update all notes and reports to reflect the actual schematic designators:
  - DS18B20 Pull-up: R61 (previously R19)
  - LED Series Resistors: R59/R60 (previously R33/R36)
  - INA333: U10 (previously U8)
- BOM Update: Correct J5/J6 descriptions from "LED_BLANK" to "LED_BANK".
- Verify and update the stated standby current in the final design notes.

Sheet: /
File: COMBO_NODE.kicad_sch
Title: TITLE_HEADER
Size: A4 | Date: 2026-10-25 | **Rev: 3.6**
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COMBO NODE SUBSYSTEM DESIGN & REVIEW NOTES (REV 3.6)

Project: OpenMycoNet COMBO_NODE
Sheet: /RESULT AND REVIEW/
Date: 2026-10-25
Revision: 3.6
Status: READY FOR MANUFACTURING (JLCPCB 4-Layer Solid GND Stackup)

1. WATCHDOG SUPERVISOR STANDARDIZATION

- Component & Value: TPS3851H33 (U7) timing capacitor C45 synchronized to 100 nF across Schematic, PCB Layout, BOM, and CPL deliverables.
- Timeout Period: Standardized to 323 ms nominal (tWD = 3.23 * 100nF + 0.381 ms) across all engineering reports, eliminating false resets during active I2C/SPI sensor acquisition cycles.

2. INA333 BIO-SIGNAL ANALOG FRONT-END PROTECTION (U10 / CASE 3)

- Signal Flow Order: Terminal J4 -> Clamping Diodes D3 (BAV199DW) -> Series Resistors R56/R57 -> INA333 (U10) Inputs.
- Dual-Stage Rail Clamping: Ultra-low-leakage diode array D3 (BAV199DW) clamps differential surge/ESD pulses directly to +3.3VA and GND prior to entering U10.
- Current Limiting: Precision series resistors R56 & R57 (1 kΩ, 0.1%) restrict residual fault currents below 5 mA.

3. OPTICAL ILLUMINATION & PROBE INTERFACE (J5/J6 – LED_BANK / POINT 10)

- Off-Board LED Drivers: On-board LEDs migrated to 2-pin screw terminals J5 and J6 (LED_BANK) for 470 nm optical probe connections.
- Rail Selection & Current Limiting: Current-limiting resistors R59 and R60 (68 Ω, 0805, 1/8 W) connected directly to +BATT supply (instead of VDD / +3.3V) to prevent digital supply sagging. Forward current fixed at ~16.2 mA @ 4.2 V BATT.
- Low-Side Switching: Controlled by BSS138 N-channel MOSFETs (Q1, Q2) driven by active-HIGH logic (LED_PWM1 / LED_PWM2 on GPIO11 / GPIO10) with 10 kΩ gate pull-down resistors (R34, R37).

4. SOIL MOISTURE POWER GATING (CASE 5)

- High-Side Load Switch: TPS22919DCKR (U18) controlled by GPIO41 (MOISTURE_SLEEP) gates the MOISTURE_VDD supply rail at J9 Pin 1, eliminating 5 mA – 15 mA sensor idle power drain during MCU deep sleep. Controlled Quick Output Discharge (QOD) discharges bulk capacitor C47 (47 µF) via R58 (1 kΩ) to GND when disabled.
- Bus Continuity: Precision ADC U14 (ADS1115) VDD remains powered on continuous +3.3VA to keep the shared I2C bus active without ESD diode back-feeding.

5. EC EXCITATION PATH & RC HARMONICS SUPPRESSION (CASE 4)

- Low-Pass Filtering: Passive RC filter R38 (1 kΩ) and C29 (10 nF) placed at U13 non-inverting input (+INA / Pin 3) establishes a 15.9 kHz cutoff frequency.
- Analog Front-End Shutdown: Buffer U13 (OPA2320S) shutdown pins (~SHDN1 / ~SHDN2) tied to ANA_SHUT (GPIO42), driving the AFE into a < 1 µA standby state in deep sleep.

6. 1.65 V MIDPOINT REFERENCE BUFFER & DISTRIBUTION (VMID_REFF / POINTS 06, 14)

- Midpoint Divider: R43/R44 standardized to 100 kΩ filtered by C34 (100 nF) to GND, providing a stable 1.65 V midpoint with minimal current draw (16.5 µA from +3.3VA).

- Standalone Voltage Follower: Actively buffered by dedicated low-quiescent single op-amp U19 (OPA333AIDBVR, SOT-23-5) configured as a 1:1 unity-gain follower. Pin 3 (+IN) connects to C34, and Pin 4 (-IN) is shorted to Pin 1 (OUT). Pin 1 drives net VMID_REFF through R62 (22 Ω isolation resistor) with local decoupling C51 (100 nF to GND).

- Stage Isolation: U13B is completely isolated from R62/VMID_REFF and reserved exclusively for the EC probe receive amplifier, preventing soil-signal crosstalk during active EC stimulation.

Star Reference Distribution: VMID_REFF cleanly distributed to:
1) INA333 (U10) Pin 5 (REF) for mid-rail bi-directional output swing.
2) Input bias offset network R32.
3) High-value 10 MΩ bias return resistors R63 and R64.
4) EC stimulation return path (J11 Pin 2 & bottom of R46).

7. ON-BOARD THERMISTOR SCHEMATIC RETENTION & DNP RATIONALE (TH1 / POINT 12)

- Manufacturing Deliverables: TH1 footprint is designated Do Not Populate (DNP) and explicitly excluded from the final Bill of Materials (BOM) and Pick & Place (CPL) export files. JLCPCB manufacturing runs will leave TH1 unpopulated.
- Primary 3-Wire Battery Topology: Standard 3-wire LiPo battery packs (connected via J2) utilize an internal 10 kΩ pack NTC. Leaving TH1 unpopulated prevents parallel resistance errors (~5 kΩ equivalent) on BQ24075 (U4) Pin 1 (TS) that would cause false thermal faults and block charging.
- Schematic Retention & Debugging: TH1 symbol is intentionally retained on the schematic drawing to document hardware fallback options. During bench testing or field deployment, if a 2-wire battery (lacking an internal NTC) is utilized, TH1 can be manually hand-soldered onto the PCB pads to enable local thermal protection without requiring a schematic or PCB revision.

8. ANALOG & DIGITAL POWER RAIL ISOLATION (FB1 REMOVAL / POINT 13)

- Topology Correction: Removed FB1 bridging resistor between +3.3V and +3.3VA.
- Independent Regulation: Buck-boost converters U2 (+3.3V) and U3 (+3.3VA) operate with complete physical and electrical isolation, preventing high-frequency digital switching noise from polluting the precision analog reference plane.

9. EC COUPLING CAPACITOR UPGRADE (C28 / POINT 07)

- Capacitance Upgrade: C28 increased to 100 µF (1206 MLCC, >= 6.3V rating).
- Impedance Optimization: Lowers 16 Hz AC capacitive reactance (Xc) down to ~99.5 Ω, eliminating signal attenuation across high-impedance liquid/soil probes.

10. ONE-WIRE TEMP SENSOR PULL-UP & GROUND CORRECTION (POINTS 08, 09, 11, 15)

- Bus Pull-Up: Integrated dedicated 4.7 kΩ pull-up resistor (R61) between DS18B20_DATA (GPIO6) and +3.3V rail to ensure reliable 1-Wire communication.
- Net Re-routing: C22 ground pin re-routed from digital GNDA to the continuous unified Layer 2 ground plane (GND).
- Local Decoupling: Localized 100 nF ceramic decoupling capacitor (C48) placed directly adjacent to OPA2320 (U13) Pin 10 (+3.3VA) and GND.

Sheet: /RESULT AND REVIEW/
File: BLOCK_DIAGRAM.kicad_sch

Title: RESULT AND REVIEW

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BLOCK DIAGRAM & SUBSYSTEM INTERCONNECT DESIGN NOTES (REV 3.6)

Project: OpenMycoNet COMBO_NODE
Status: READY FOR MANUFACTURING (JLCPCB 4-Layer Stackup)
Date: 2026-10-25
Revision: 3.6

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1. HIERARCHICAL BUS ARCHITECTURE & INTERCONNECT MATRIX

* Digital Core & Communication Buses:
  * USB DATA Bus: Native USB 2.0 (`USB_D+`, `USB_D-`) routed from USB-C connector J1 (Sheet 4) through USBLC6-2SC6 ESD protection array U1 and 27 Ω series resistors R3/R4 directly to ESP32-S3 USB transceiver (Sheet 7).
  * Shared I2C Bus: Shared I2C BUS (`SDA`, `SCL`) pulled up via 4.7 kΩ resistors R17/R18 on MCU sheet, interfacing MAX17048 Fuel Gauge U5 (Sheet 6), ADS1115 ADC U14 (Sheet 8), DS3231 RTC U10 (Sheet 8), and SCD41 CO2 Sensor U11 (Sheet 8).
  * Shared SPI Bus: High-speed SPI (`SPI_CS`, `SPI_MOSI`, `SPI_SCK`, `SPI_MISO`) driving SX1262 LoRa Transceiver U16 (Sheet 9) and Hirose DM3AT MicroSD Card Interface J13 (Sheet 10).
  * Debug UART0 Bus: `TXD0` and `RXD0` lines from ESP32-S3 routed directly to Expansion Header J3 (Pins 8 & 10) for firmware flashing and real-time terminal output.
* Analog Signal & Excitation Control Nets:
  * `DAC_OUT` / `DAC_STIM`: 16 Hz AC stimulation line generated by MCU DAC (GPIO17/Pin 38), filtered via RC low-pass network (R38 = 1 kΩ, C29 = 10 nF relocated directly to U13 Pin 3 +INA), and routed through selectable OPA2320S buffer stage U13.
  * `EC_ENABLE`: Dedicated digital line from ESP32-S3 (GPIO48/Pin 25) toggling TS5A3159 analog switch U15 for dual-mode biological impedance and excitation path selection.
  * `BIO_IN+` / `BIO_IN-`: Differential microvolt bio-potential sensing lines from J4 connector routed through R56/R57 1 kΩ series current-limiting resistors into BAV199DW clamping array D3 and INA333 instrumentation amplifier U10.
  * `MOISTURE_ANALOG`: Analog output from Soil Moisture Connector J9 routed to ADS1115 ADC U14 Channel AIN0 (Pin 4).

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2. POWER OPTIMIZATION & DEEP SLEEP CONTROL NETS

* `ANA_SHUT` (Analog Front-End Shutdown):
  * Hierarchical Signal: Output from ESP32-S3 (GPIO42, Pin 35) routed directly to ~SHDN1 (Pin 5) and ~SHDN2 (Pin 10) of U13 (OPA2320SAIDGSR).
  * Standby State: Pulled to GNDA via 10 kΩ resistor R23 on MCU sheet, forcing the analog front-end into ultra-low-power standby (< 1 µA) during deep sleep.
* MicroSD & Watchdog Power Gating:
  * `SD_CARD_SLEEP` / `WCD_SLEEP`: Driven by ESP32-S3 GPIO47 (Pin 34) controlling U17 (TPS22919) high-side load switch and TPS3851 SET1 pin.
  * Bus Leakage Protection: All SPI bus pull-ups (R48, R49, R51, R52) re-routed directly to the gated `SD_VDD` rail to eliminate parasitic back-feeding when `SD_VDD` is switched off.
* Soil Moisture Sensor Power Gating:
  * `MOISTURE_SLEEP`: Driven by ESP32-S3 GPIO41 (Pin 34) controlling U18 (TPS22919) high-side load switch.
  * Gated Rail Routing: Gates power rail `MOISTURE_VDD` directly to Pin 1 of J9 terminal, eliminating continuous 5 mA – 15 mA sensor idle drain during sleep.
  * ADC Power Isolation: ADS1115 ADC U14 VDD (Pin 8) remains tied to continuous clean analog power `+3.3VA` to keep the shared I2C bus active without back-feeding through internal ESD diodes.
* Static Power LED Elimination:
  * Removed static PCB power indicator LEDs on +BATT and VDD supply rails, saving ~7.7 mA continuous quiescent power loss.

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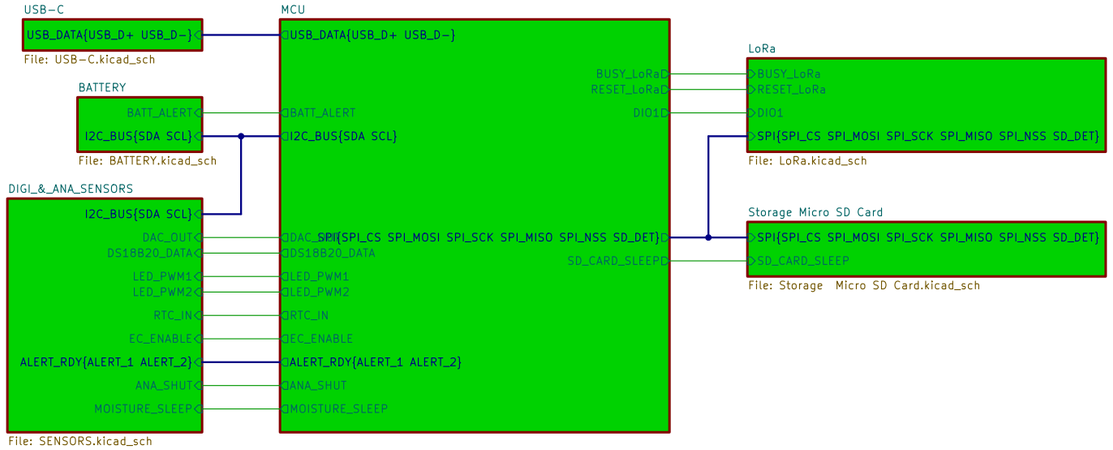
3. SYSTEM SAFETY, BATTERY & PROTECTION ARCHITECTURE

* Charger & Power-Path Manager Configuration:
  * Core IC: BQ24075 (U4) standalone Li-ion power-path manager on Sheet 6.
  * Mode Setup: EN2 pin (Pin 5) pulled HIGH to VBUS via 10 kΩ resistor R12, and EN1 (Pin 6) grounded. Guarantees stable VDD system power regulation during batteryless USB operation.
  * Power Switch S1: Hardware slide switch connected to SYSOFF (Pin 15) to toggle system power ON/OFF.
* External Battery Thermal Protection:
  * TH1 NTC Status: On-board thermistor TH1 designated Do Not Populate (DNP) and removed from BOM/CPL to prevent parallel 10 kΩ resistance interference (5 kΩ equivalent).
  * NTC Routing: BQ24075 TS pin (Pin 1) relies 100% on the external battery pack NTC connected through J2 Pin 2.
* Bio-Front-End Input Protection:
  * `INA333` Clamping: INA+ and INA- differential inputs protected by 1 kΩ (0.1%) series current-limiting resistors R56 and R57 placed in front of BAV199DW (D3) ultra-low-leakage rail-clamping diodes to absorb high-voltage ESD surges safely.
* Watchdog Supervisor Timing:
  * TPS3851H33 Window Watchdog U7: Driven by MCU `RST_WDI` (GPIO4/Pin 4).
  * Reset Timeout: Hardware timer configured via 100 nF capacitor C45 for a nominal 323.4 ms watchdog timeout window (t_WD = 323.4 ms).
  * Sleep Override: GPIO47 (`WCD_SLEEP`) pulls SET1 LOW prior to deep sleep to cleanly disable watchdog resets.

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4. SUBSYSTEM INTERCONNECT & PERIPHERAL ISOLATION

* Off-Board Blue Light LEDs:
  * Interface: Connectors J5 and J6 configured as 2-pin Screw Terminals.
  * Topology: Pin 1 connects to BATT+ power rail via 68 Ω series resistors (R33, R36).. Pin 2 connects to the Drain of low-side BSS138 N-channel MOSFETs Q1 and Q2.
  * Drive Logic: Active-HIGH PWM drive via `LED_PWM1` (GPIO11/Pin 18) and `LED_PWM2` (GPIO10/Pin 17) with 10 kΩ gate pull-down resistors R59/R60.
* EC Excitation Low-Pass Filter:
  * RC Network: Filter capacitor C29 (10 nF) located directly at the non-inverting input (+INA / Pin 3) of U13 (OPA2320S) after series resistor R38 (1 kΩ).
  * Frequency Cutoff



REGULATOR

File: REGULATOR.kicad_sch

Sheet: /BLOCK_DIAGRAM/
File: RESULT_&_REVIEW.kicad_sch

Title: BLOCK_DIAGRAM

Size: A3	Date: 2026-10-25	**Rev: 3.6**
KiCad E.D.A. 10.0.7		Id: 3/10

USB-C INTERFACE DESIGN NOTES (REVISED V3.6)
Project: OpenMycoNet – COMBO_NODE
Status: READY FOR MANUFACTURING (JLCPCB 4-Layer Stackup)
Date: 2026-10-25
Revision: 3.6

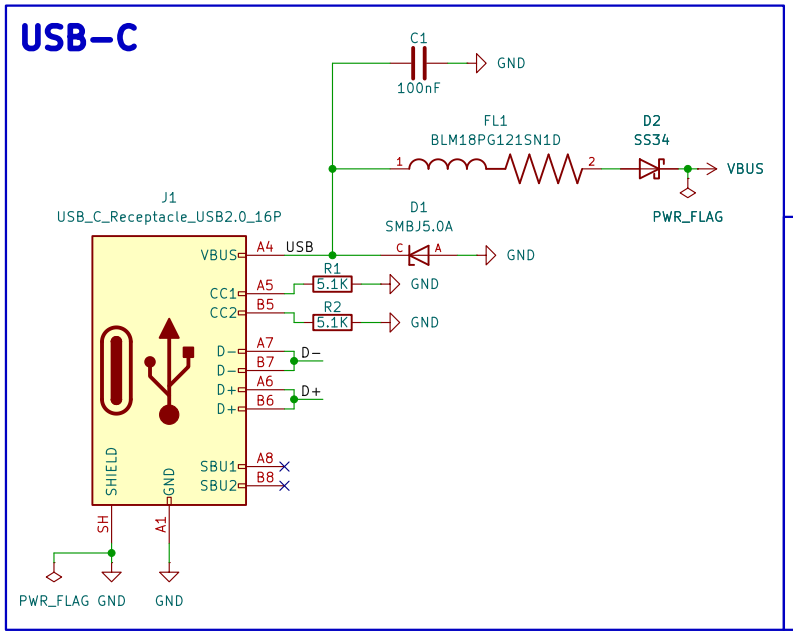
SYSTEM TOPOLOGY
Connector: J1 (USB-C 16-Pin Receptacle, USB 2.0 Full-Speed Only).
Configuration: Upstream Facing Port (UFP) device mode.
ESD Protection Array: U1 (USBLC6-2SC6) protecting data lines against electrostatic discharge.
Data Bus Routing: D+ and D- data lines routed to ESP32-S3 native USB pins via U1 ESD clamp and 27Ω series resistors R3 and R4.

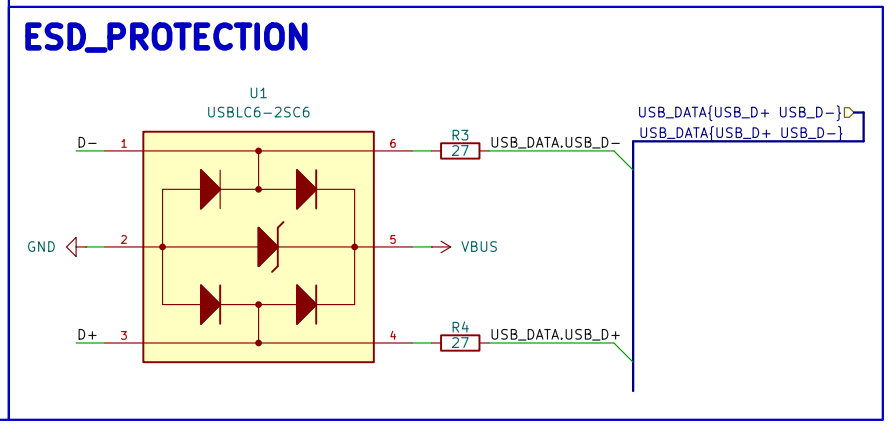
1. POWER & VBUS MANAGEMENT
Backpower Protection: D2 (SS34 Schottky Diode) prevents back-feeding power into the USB host.
EMI Filtering: FL1 ferrite bead (BLM18PG121SN1D) provides high-frequency noise filtering on VBUS input.
Input Decoupling: C1 (100nF) decouples the VBUS power rail.
Power Net Routing: Filtered VBUS net feeds directly into BQ24075 charger input and provides 10kΩ pull-up reference to EN2 pin (U4) for batteryless USB operation.

2. CC LINE CONFIGURATION
Upstream Detection: CC1 (Pin A5) and CC2 (Pin B5) pulled to GND via 5.1kΩ resistors R1 and R2.
Power Delivery: Configures Standard Downstream Port (SDP) 5V detection without requiring active USB-PD negotiation.

3. SURGE & ESD PROTECTION
VBUS TVS Clamp: D1 (SMBJ5.0A) provides IEC 61000-4-2 Level 4 surge protection on VBUS.
Data Line ESD Array: U1 (USBLC6-2SC6) protects D+ and D- lines against electrostatic discharge transients.
Series Impedance Matching: R3 and R4 (27Ω) series resistors provide impedance matching and EMI suppression.

4. COMPLETE INTERFACE MATRIX
VBUS: 5V power input rail filtered via FL1/D2 and routed directly to BQ24075 charger input and EN2 mode pin.
USB_DATA.USB_D+: Native USB D+ connected to MCU via U1 ESD array (Pin 3/4) and R4 (27Ω).
USB_DATA.USB_D-: Native USB D- connected to MCU via U1 ESD array (Pin 1/6) and R3 (27Ω).
CC1/CC2: 5.1kΩ pull-down resistors R1 and R2 for USB-C cable orientation and connection detection.





Sheet: /BLOCK_DIAGRAM/USB-C/
File: USB-C.kicad_sch

Title: USB-C

Size: A4	Date: 2026-10-25	**Rev: 3.6**
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POWER SUPPLY SUBSYSTEM DESIGN NOTES (REGULATOR BLOCK)
Project: OpenMycoNet – COMBO_NODE
Status: READY FOR MANUFACTURING (JLCPCB 4-Layer Stackup)
Date: 2026-10-25
Revision: 3.6

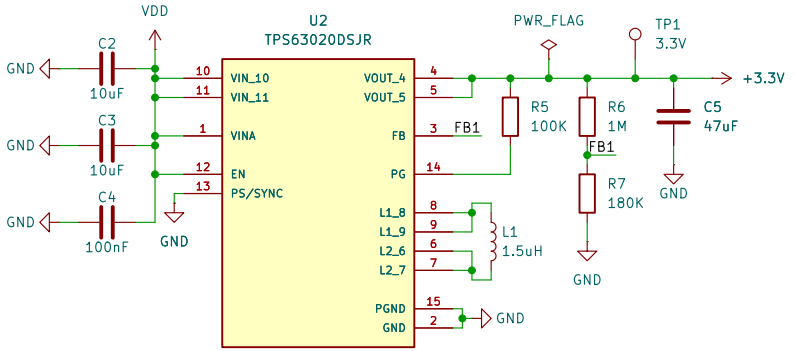
SYSTEM OVERVIEW
The power supply subsystem utilizes two independent TPS63020DSJR buck-boost converters to separate the digital core from sensitive analog measurement blocks.
Dual converters prevent high-frequency digital switching noise and transient load ripples from degrading precision sensor readings and ADC reference baselines.

1. DIGITAL POWER RAIL SETUP (+3.3V / +3.3VD)
Converter IC: U2 (TPS63020DSJR high-efficiency buck-boost converter).
Function: Regulates variable VDD input (USB 5V or LiPo battery 3.0V-4.2V) to a stable +3.3V digital supply rail.
Input Decoupling: C2 (10µF), C3 (10µF), and C4 (100nF) MLCC ceramic capacitors placed close to pins VIN_10, VIN_11, and VINA.
Feedback Network: Voltage divider using R6 (1MΩ) and R7 (180kΩ) tied to pin 3 (FB) sets the output voltage regulation to exact +3.3V (VFB = 0.5V, VOUT = 0.5V * (1 + 1M/180k) = 3.278V ~ 3.3V).
Output Filtering: Main digital output is filtered by C5 (47µF) bulk ceramic capacitor.
Power Good Monitor: Pin 14 (PG) pulled up to VOUT through R5 (100kΩ).
Power Inductor: L1 (1.5µH) connected across switching terminals L1 (pins 8, 9) and L2 (pins 6, 7).
Control Pins: EN (pin 12) tied to VDD and PS/SYNC (pin 13) grounded for continuous forced PWM active-mode operation.

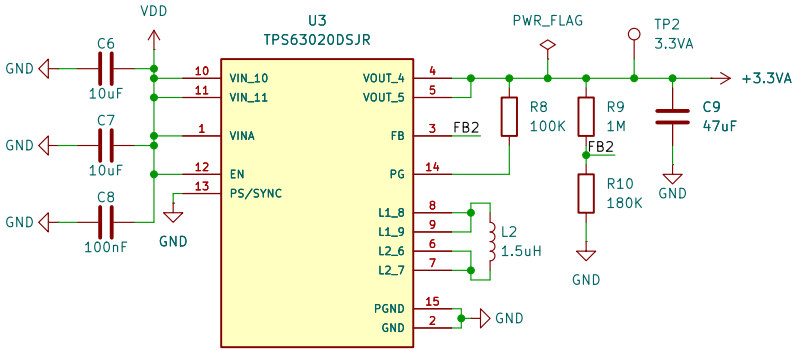
2. CLEAN ANALOG POWER RAIL SETUP (+3.3VA)
Converter IC: U3 (TPS63020DSJR high-efficiency buck-boost converter).
Function: Regulates variable VDD input into an isolated, ultra-clean +3.3VA supply rail for the analog front-end (OPA2320S, INA333, ADS1115 ADCs).
Input Decoupling: C6 (10µF), C7 (10µF), and C8 (100nF) ceramic capacitors connected across VIN_10, VIN_11, and VINA.
Feedback Network: Voltage divider using R9 (1MΩ) and R10 (180kΩ) tied to pin 3 (FB) sets the analog output rail regulation to +3.3VA.
Output Filtering: Low-pass output filtering provided by C9 (47µF) bulk ceramic capacitor.
Power Good Monitor: Pin 14 (PG) pulled up to VOUT through R8 (100kΩ).
Power Inductor: L2 (1.5µH) connected across switching terminals L1 (pins 8, 9) and L2 (pins 6, 7).
Control Pins: EN (pin 12) tied to VDD and PS/SYNC (pin 13) grounded for low-noise continuous mode operation.

3. GROUND PLANE & RULE VERIFICATION SETUP
Common System Ground (GND): Solid common ground return plane utilized across all 4 layers under Revision 3.3 to eliminate loop antennas and return-path discontinuities.
PWR_FLAG: Tied directly to the power rails and GND nets on both U2 and U3 output domains for automated KiCad Electrical Rules Check (ERC) power-driver verification.

BUCK CONVERTER: OUT ~ 3.3V



BUCK CONVERTER: OUT ~ 3.3VA



Sheet: /BLOCK_DIAGRAM/REGULATOR/
File: REGULATOR.kicad_sch

Title: REGULATOR

Size: A4	Date: 2026-10-25	**Rev: 3.6**
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BATTERY SUBSYSTEM DESIGN NOTES (BATTERY BLOCK)
Project: OpenMycoNet – COMBO_NODE
Status: READY FOR MANUFACTURING (JLCPCB 4-Layer Stackup)
Date: 2026-10-25
Revision: 3.6

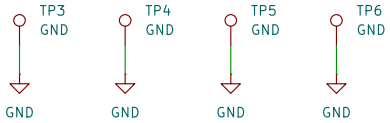
SYSTEM OVERVIEW
The battery subsystem manages dynamic power-path routing, lithium battery charging, hardware power switching, state-of-charge fuel gauging,
\and edge ground probing. It utilizes a BQ24075 power-path manager (U4) to seamlessly power downstream regulators (VDD)
from either USB Input (VBUS) or a single-cell LiPo/Li-Ion battery (+BATT). A MAX17048 fuel gauge (U5) provides battery telemetry over I2C,
while a dedicated slide switch (S1) controls physical battery isolation.

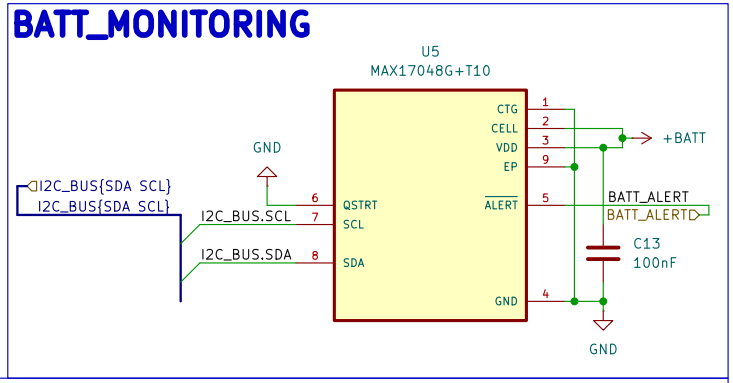
1. BATTERY CHARGER, POWER-PATH & POWER SWITCH (U4 BQ24075RGT)
* Core IC: U4 (BQ24075RGT standalone Li-ion power-path manager).
* Hardware Power Switch (S1): Slide switch S1 connected to pin 15 (SYSOFF). Toggling S1 switches SYSOFF between GND
(normal battery operation / power ON) and +BATT (battery disconnect / system OFF).
* Input Mode Configuration (EN1/EN2): Pin 5 (EN2) is pulled HIGH to VBUS through a 10kΩ resistor (R12), and Pin 6 (EN1) is tied LOW to GND.
Per Table 1 of the BQ24075 datasheet, this EN2=1, EN1=0 state configures External Resistor Mode via pin 12 (ILIM)
and ensures stable VDD passthrough during batteryless USB operation.
* Input Current Limit: Resistor R13 (1.18kΩ) on pin 12 (ILIM) configures the programmable maximum input current draw from VBUS to 1.35A.
* Fast-Charge Current: Resistor R14 (1.13kΩ) on pin 16 (ISET) sets fast-charge current to 787mA.
* Input Decoupling: C10 (1µF) ceramic capacitor connected between VBUS input and GND.
* Battery Terminal Connection: BAT pin (pin 2) connects to the single-cell battery rail +BATT, decoupled by C12 (4.7µF).
Connected to 3-pin battery connector J2 (BATT_CONN).
* System Power Rail (VDD): OUT pin (pin 10) provides the main system output rail VDD, decoupled by C11 (4.7µF) and monitored by test point TP7.
* Charge Enable: Pin 4 (~CE) pulled LOW to GND to enable automatic charging when VBUS is present.
* Safety Timer: Pin 14 (TMR) tied to GND through R11 (46.4kΩ) to set safety timer limits.
* Thermal Protection: Pin 1 (TS) tied to 10kΩ NTC thermistor TH1 on PCB as well as Pin 2 (NTC) of 3-pin J2 header
to suspend charging during over/under-temperature conditions. TH1 is designated DNP when using external pack NTC.
* Status Monitors: PGOOD (pin 7) pulled up to +3.3V via R15 (10kΩ). CHG (pin 9) pulled up to +3.3V via R16 (10kΩ).
Static PCB power/alert LEDs (D3/D4) removed to eliminate continuous parasitic drain during deep sleep.

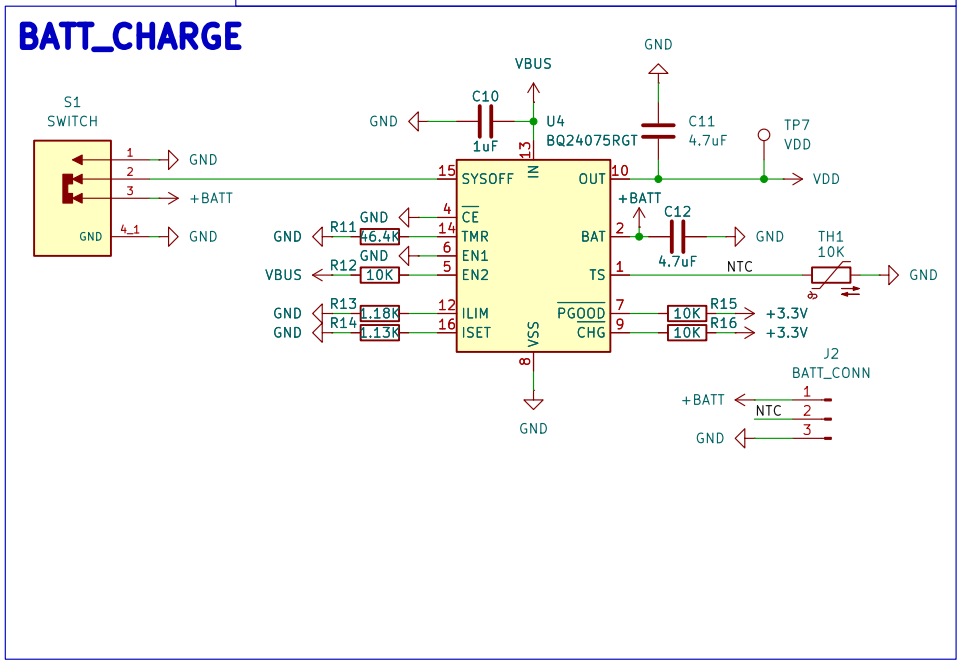
2. FUEL GAUGE SETUP (U5 MAX17048G+T10)
* Core IC: U5 (MAX17048G+T10 ModelGauge I2C fuel gauge).
* Sensing & Power: CELL (pin 2) and VDD (pin 3) tied directly to the +BATT rail, decoupled by C13 (100nF).
* Grounding: GND (pin 4), CTG (pin 1), and Exposed Pad EP (pin 9) connected directly to system GND.
* I2C Telemetry: SCL (pin 7) and SDA (pin 8) routed to the shared I2C bus (I2C_BUS.SCL, I2C_BUS.SDA).
* Interrupt Output: ALERT (pin 5) drives active-LOW BATT_ALERT signal directly to MCU GPIO21 for low-power wake-up triggers.
* Quick-Start Pin: QSTRT (pin 6) tied to GND.

3. PCB EDGE GROUNDING & TEST POINT ARRAY
* Edge Ground Test Points: Dedicated test points TP3, TP4, TP5, and TP6 are placed at the perimeter/corners of the schematic layout.
* Function: Provides reliable multimeter/oscilloscope ground clip attachment points across the PCB boundary
and facilitates chassis grounding checks without cluttering internal signal lines.

4. INTERFACE MATRIX
* VBUS: +5V power input supplied from the USB-C interface.
* VDD: Regulated system power output driving downstream buck-boost converters (+3.3V / +3.3VA).
* +BATT: Unregulated single-cell LiPo/Li-ion battery rail (3.0V to 4.2V nominal).
* S1 (SWITCH): Hardware system power switch (toggles SYSOFF).
* J2 (BATT_CONN): 3-pin header (+BATT, NTC, GND) for external battery pack interfacing.
* TP3–TP6: PCB perimeter Ground (GND) test points.







Sheet: /BLOCK_DIAGRAM/BATTERY/
File: BATTERY.kicad_sch

Title: BATTERY

Size: A4	Date: 2026-10-25	**Rev: 3.6**
KiCad E.D.A. 10.0.7		Id: 6/10

MCU SUBSYSTEM DESIGN NOTES (MCU BLOCK)

Project: OpenMycoNet COMBO_NODE
Status: READY FOR MANUFACTURING (JLCPCB 4-Layer Stackup)
Date: 2026-10-25
Revision: 3.6

SYSTEM OVERVIEW

The MCU subsystem centers on the ESP32-S3-WROOM-1-N16R8 module (U6) operating from a regulated +3.3V digital supply rail. It controls real-time sensor sampling, dual-mode analog front-end multiplexing, SPI/I2C communication, power gating load switches, and external window watchdog supervision.

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1. SYSTEM CORE AND POWER

* Core Module: U6 (ESP32-S3-WROOM-1-N16R8).
* Power Rail Filtering: Parallel ceramic decoupling capacitors connected directly across +3.3V (Pin 2) to GND (Pin 1).
* Hardware Reset: Active-LOW ~RESET line on EN (Pin 3) pulled HIGH to +3.3V via R27 (10kΩ) and debounced by C18 (100nF) via SW1 push button. Driven by watchdog supervisor U7 output (~RESET).
* Hardware Watchdog Supervisor: U7 (TPS3851H33EDDAR) monitors system software execution via RST_WDI (IO4 / Pin 4). If no heartbeat pulse is toggled within the 323.4 ms window (C45 = 100nF), U7 asserts ~RESET_CHIP LOW to perform a hard chip reset.
* Sleep Mode Watchdog Disable: WCD_SLEEP (IO47 / Pin 24) drives U7 SET1 (Pin 5). Driving IO47 LOW before entering deep sleep cleanly disables the watchdog supervisor.

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2. BOOT, STRAPPING AND MEMORY CONFIGURATION

* Bootloader Download Mode: IO0 (Pin 27) connected to BOOT0 net, pulled HIGH to +3.3V via R28 (10kΩ) and grounded via SW2 push button (debounced by C19 = 100nF).
* ROM Log Suppression Strapping: IO45 (Pin 26) and IO46 (Pin 16) pulled LOW to GND via R24 (10kΩ) and R25 (10kΩ) respectively to suppress internal ROM boot debug messages.
* Octal PSRAM Protection: IO35 (Pin 28), IO36 (Pin 29), and IO37 (Pin 30) explicitly unassigned (NC) to avoid memory bus conflicts with internal Octal PSRAM routing.

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3. BUS PULL-UPS AND CONTROL LOGIC

* Battery Interrupt: IO21 (Pin 23) receives active-LOW BATT_ALERT signal from fuel gauge, pulled HIGH to +3.3V via R20 (10kΩ).
* Shared I2C Bus Pull-Ups: I2C_BUS.SCL (IO8 / Pin 12) and I2C_BUS.SDA (IO9 / Pin 17) pulled HIGH to +3.3V via resistors R17 (4.7kΩ) and R18 (4.7kΩ).
* Analog Front-End Shutdown Control: IO42 (Pin 35) outputs ANA_SHUT net to control OPA2320S shutdown pins (~SHDN1/~SHDN2), pulled LOW to GND via R23 (10kΩ).
* Dual-Mode Analog Switch Control: IO48 (Pin 25) outputs EC_ENABLE net to toggle TS5A3159 analog switch path, pulled LOW to GND via R26 (10kΩ).
* Soil Moisture Power Control: IO41 (Pin 34) outputs MOISTURE_SLEEP net to drive U18 (TPS22919) load switch, gating power to off-board moisture probes.
* MicroSD Power Control: IO40 (Pin 33) outputs SD_CARD_SLEEP net to drive U17 (TPS22919) load switch.
* ADC Interrupt Pull-Ups: IO38 (Pin 31, ALERT_RDY_ALERT_1) and IO39 (Pin 32, ALERT_RDY_ALERT_2) pulled HIGH to +3.3V via R21 (10kΩ) and R22 (10kΩ) to support open-drain ADS1115 interrupt outputs.

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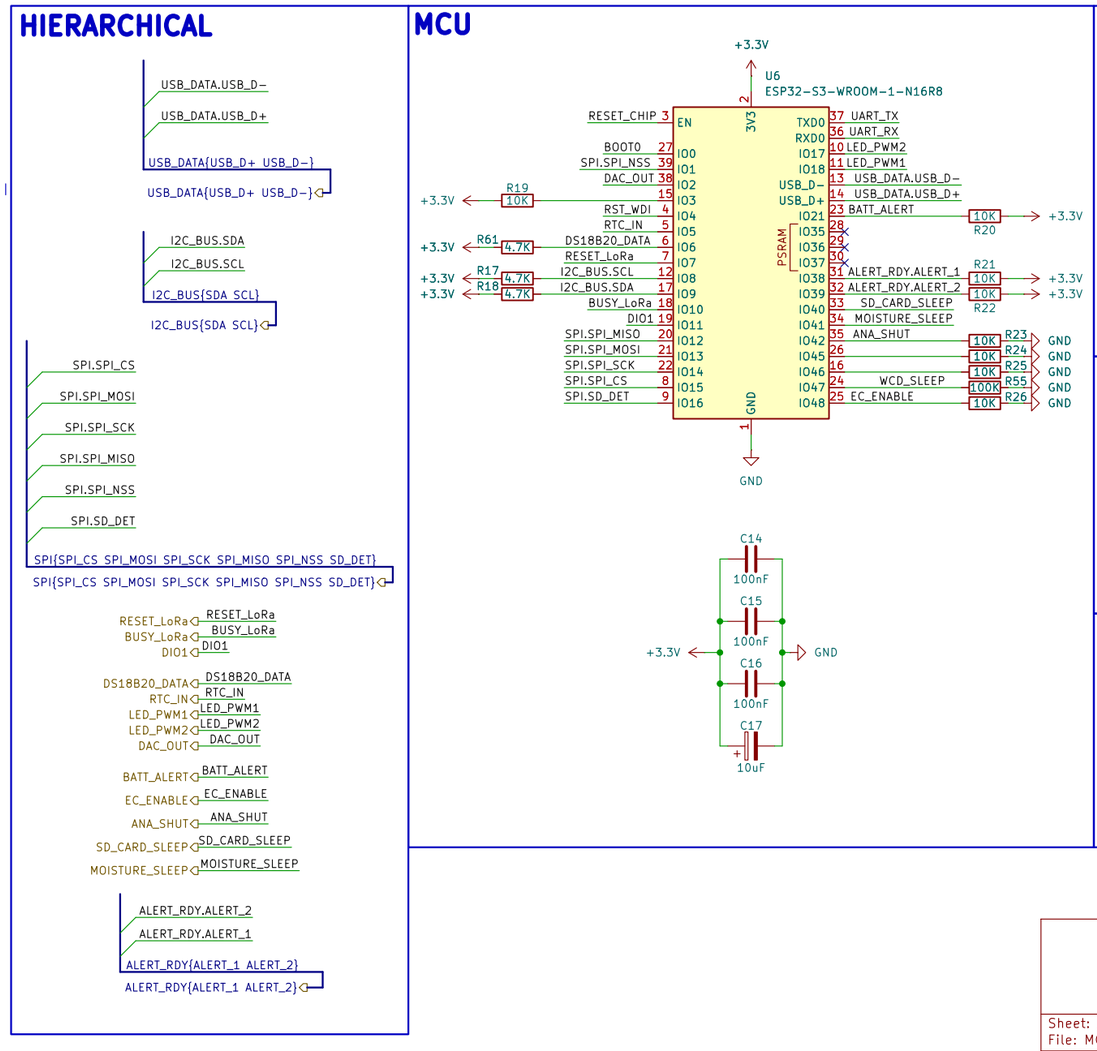
4. MASTER GPIO ASSIGNMENT TABLE (REV 3.5 EXACT SCHEMATIC MATCH)

PIN#	GPIO / PIN NAME	NET NAME / SIGNAL	FUNCTION & CONFIGURATION DESCRIPTION
1	GND	GND	System ground reference connection
2	3V3	+3.3V	Main +3.3V system power supply rail
3	EN	RESET_CHIP	System hardware reset line (Pulled HIGH via R27, connected to U7 ~RESET)
27	IO0	BOOT0	Bootloader download mode select (Pulled HIGH via R28, debounced by SW2/C19)
39	IO1	SPI.SPI_NSS	LoRa Transceiver Chip Select line
38	IO2	DAC_OUT	16 Hz AC excitation output line for bio-potential sensor
15	IO3	NC	Isolated strapping pin (Pulled HIGH to +3.3V via R19 10kΩ)
4	IO4	RST_WDI	TPS3851H33 Watchdog WDI input pulse line (Toggled during execution)
5	IO5	RTC_IN	DS3231MZ real-time clock alarm interrupt input
6	IO6	DS18B20_DATA	DS18B20 1-wire temperature sensor data bus (Pulled HIGH to +3.3V via R61 4.7kΩ)
7	IO7	RESET_LoRa	Active-LOW hardware reset line for SX1262 LoRa transceiver
12	IO8	I2C_BUS.SCL	Shared I2C bus clock line (Pulled HIGH via R17 4.7kΩ)
17	IO9	I2C_BUS.SDA	Shared I2C bus data line (Pulled HIGH via R18 4.7kΩ)
18	IO10	BUSY_LoRa	SX1262 LoRa internal status indicator line
19	IO11	DIO1	SX1262 LoRa interrupt request line
20	IO12	SPI.SPI_MISO	Shared SPI bus Master-In Slave-Out data line
21	IO13	SPI.SPI_MOSI	Shared SPI bus Master-Out Slave-In data line
22	IO14	SPI.SPI_SCK	Shared SPI bus serial clock line
8	IO15	SPI.SPI_CS	Active-LOW MicroSD chip select line
9	IO16	SPI.SD_DET	MicroSD card detect input line
10	IO17	LED_PWM2	PWM gate drive for LED MOSFET driver Q2
11	IO18	LED_PWM1	PWM gate drive for LED MOSFET driver Q1
13	USB_D-	USB_DATA.USB_D-	Native USB 2.0 full-speed data minus line
14	USB_D+	USB_DATA.USB_D+	Native USB 2.0 full-speed data plus line
23	IO21	BATT_ALERT	Active-LOW fuel gauge state-of-charge interrupt (Pulled HIGH via R20)
28	IO35	NC	Reserved for internal Octal PSRAM (Unconnected / No Connect)
29	IO36	NC	Reserved for internal Octal PSRAM (Unconnected / No Connect)
30	IO37	NC	Reserved for internal Octal PSRAM (Unconnected / No Connect)
31	IO38	ALERT_RDY.ALERT_1	ADS1115 #1 hardware interrupt/wake-up line (Pulled HIGH via R21)
32	IO39	ALERT_RDY.ALERT_2	ADS1115 #2 hardware interrupt/wake-up line (Pulled HIGH via R22)
33	IO40	SD_CARD_SLEEP	High-side load switch enable for MicroSD (U17)
34	IO41	MOISTURE_SLEEP	High-side load switch enable for Soil Moisture (U18)
35	IO42	ANA_SHUT	Analog front-end shutdown control line (Pulled LOW via R23 10kΩ)
26	IO45	GND (Strapping)	Boot strapping pin (Pulled LOW to GND via R24 10kΩ)
16	IO46	GND (Strapping)	Boot strapping pin (Pulled LOW to GND via R25 10kΩ)
24	IO47	WCD_SLEEP	Watchdog enable control line (U7 SET1 pin, Pulled LOW via R55 100kΩ)
25	IO48	EC_ENABLE	Digital control line for analog switch (Pulled LOW via R26 10kΩ)
37	TXD0	UART_TX	Debug hardware UART0 transmit line
36	RXD0	UART_RX	Debug hardware UART0 receive line

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5. EXPANSION HEADER J3 INTERFACE

* Pin 1: I2C_BUS.SCL (Shared I2C clock line from IO8 / Pin 12)
* Pin 2: I2C_BUS.SDA (Shared I2C data line from IO9 / Pin 17)
* Pin 3: SPI.SPI_MOSI (Shared SPI Master-Out Slave-In from IO13 / Pin 21)
* Pin 4: SPI.SPI_SCK (Shared SPI clock from IO14 / Pin 22)
* Pin 5: SPI.SPI_MISO (Shared SPI Master-In Slave-Out from IO12 / Pin 20)
* Pin 6: GND (Ground reference connection)
* Pin 7: +3.3V (Digital system 3.3V supply)
* Pin 8: UART_RX (Hardware UART0 receive on RXD0 / Pin 36)
* Pin 9: +3.3V (Digital system 3.3V supply)
* Pin 10: UART_TX (Hardware UART0 transmit on TXD0 / Pin 37)



Sheet: /BLOCK_DIAGRAM/MCU/
File: MCU.kicad_sch

Title: MCU

Size: A3	Date: 2026-10-25	**Rev: 3.6**
KiCad E.D.A. 10.0.7		Id: 7/10

DIGITAL & ANALOG SENSORS SUBSYSTEM DESIGN NOTES (SENSORS BLOCK)
==========
Project: OpenMycoNet COMBO_NODE Rev 3.6
Status: FINAL PRODUCTION RELEASE

1. BIOELECTRIC & IMPEDANCE FRONT-END (ELECTRODES & EC STIMULATION)

• INA333 INSTRUMENTATION AMPLIFIER (U10):
  – Measures microvolt-level bioelectric potential differences between INA+ and INA– inputs.
  – ESD and transient protection provided by inline 10 kΩ series resistors (R56, R57) and low-capacitance ESD clamp array (D3).
  – Pin 5 (REF) tied directly to active VMID_REFF (1.65 V) to center output swing at mid-rail.

• 1.65 V MIDPOINT REFERENCE BUFFER & DISTRIBUTION (U19 / VMID_REFF):
  – Midpoint Divider: R43/R44 standardized to 100 kΩ filtered by C34 (100 nF) to GND, providing a stable 1.65 V midpoint with minimal current draw (16.5 µA).
  – Standalone Voltage Follower: Actively buffered by dedicated low-quiescent op-amp U19 (OPA333AIDBVR, SOT-23-5) with Pin 3 (+IN) tied to C34, and Pin 4 (–IN) shorted to Pin 1 (OUT) in a 1:1 unity-gain configuration.
  – Active Output: Pin 1 drives net VMID_REFF through R62 (22 Ω isolation resistor) with dedicated local decoupling C51 (100 nF to GND).
  – Star Reference Distribution: VMID_REFF provides clean 1.65 V virtual ground to:
    1) INA333 (U10) Pin 5 (REF) for mid-rail bi-directional output swing.
    2) Input bias offset network R32.
    3) High-value 10 MΩ bias return resistors R63 and R64.
    4) EC stimulation return path (J11 Pin 2 & bottom of R46).

• OPA2320 TRANSIMPEDANCE / EC RECEIVER (U13):
  – Channel A (U13A): Functions as high-gain transimpedance stage for bio-impedance signals.
  – Channel B (U13B): Serves strictly as the EC probe receive amplifier. Pin 7 (+INB) connects to 1.65 V midpoint, and Pin 9 (VOUT_B) feeds back to Pin 8 (INB–) through parallel network R40||R41 = 1.65 kΩ and C30 = 10 pF. Completely isolated from R62/VMID_REFF to prevent soil signal crosstalk during active EC stimulation.
  – Shutdown Control: ANA_SHUT line from MCU GPIO42 controls active/sleep modes (< 1 µA sleep).

• TS5A3159 ANALOG SWITCH & ATTENUATOR (U15):
  – Provides gain-switching and AC excitation path selection for EC probe measurements.
  – Driven by EC_ENABLE via MCU GPIO48 (pulled LOW by R26 to disable stimulus at boot).

2. DIGITAL SENSOR ARRAY & ANALOG CONVERSION

• ADS1115 16-BIT PRECISION ADC (U14):
  – Digitizes low-frequency analog sensor outputs over shared I2C bus (SDA/SCL).
  – Powered via continuous +3.3VA rail for noise-free conversion baseline.

• SCD41 NDIR CO2 SENSOR (U11):
  – Measures ambient CO2, temperature, and relative humidity over shared I2C bus.
  – Software low-power commands utilized to enter low-power single-shot measurement mode.

• DS3231MZ HIGH-PRECISION RTC (U12):
  – Provides precise timestamping; backed up by CR1216 coin cell (BT1) on VBATT rail.
  – Communicates via I2C with dedicated interrupt pin connected to MCU.

• DS18B20 1-WIRE DIGITAL TEMPERATURE PROBE (J8):
  – Dedicated 1-Wire interface with 4.7 kΩ pull-up resistor (R61) to +3.3V.

3. POWER-GATED SENSORS & ILLUMINATION PROBES

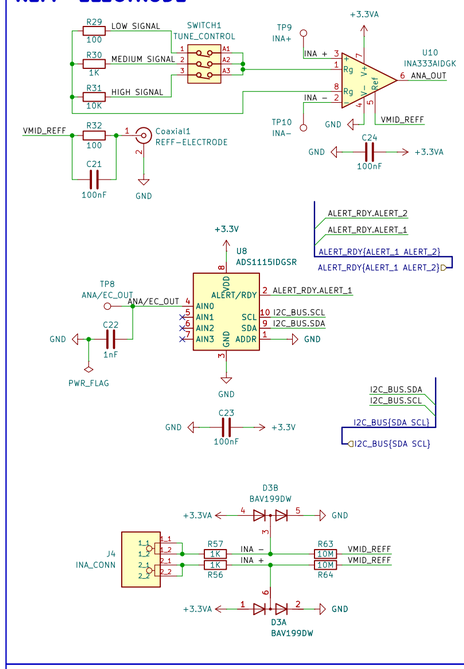
• TPS22919 LOAD SWITCH (U18 – MOISTURE SENSOR POWER GATE):
  – Controls power to capacitive soil moisture probe (J9) via MOISTURE_SLEEP (GPIO41).
  – Completely isolates moisture sensor power rail during MCU deep sleep to extend battery life.

• 470 nm BLUE LIGHT ILLUMINATION PROBES (J5 / J6 – LED BANK):
  – Pin 1 (Power): Connected directly to +BATT supply through dedicated 68 Ω series resistors (R59, R60, 0805, 1/8 W). Forward current fixed at ~16.2 mA @ 4.2 V BATT (VF ≈ 3.1 V), preventing voltage sags on the primary +3.3V digital supply.
  – Pin 2 (Control): Connected to Drain of low-side BSS138 N-channel MOSFETs (Q1, Q2). driven in active-HIGH logic via LED_PWM1 (GPIO11) and LED_PWM2 (GPIO10) with 10 kΩ gate pull-downs (R34, R37) to ensure zero emission during MCU boot.

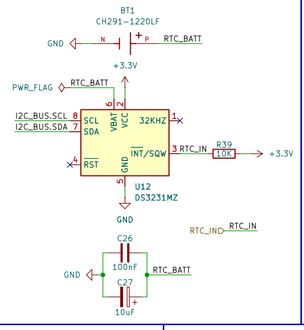
4. SUBSYSTEM INTERCONNECT MATRIX

COMPONENT / CONNECTOR	SIGNAL / NET NAME	INTERFACE TYPE	POWER RAIL / CONTROL
U10 (INA333)	INA+ / INA–	Analog Diff	+3.3VA / R56/57+D3 Prot.
U13 (OPA2320S)	ANA_SHUT	Digital GPIO42	+3.3VA (Sleep < 1 µA)
U19 (LP9811DBV)	VMID_REFF	Analog Ref	+3.3VA / Standby 425 nA
U15 (TS5A3159)	EC_ENABLE	Digital GPIO48	+3.3VA (Pulled LOW R26)
EC PROBE DRIVER	DAC_OUT	16 Hz AC Stim	+3.3VA / C28 (100 µF)
U14 (ADS1115)	I2C_BUS (SDA/SCL)	Shared I2C	+3.3VA (Continuous)
U11 (SCD41)	I2C_BUS (SDA/SCL)	Shared I2C	Soft-Sleep Command
DS3231MZ (RTC)	RTC_IN	I2C / INT	+3.3V / CR1216 Backup
J8 (DS18B20)	DS18B20_DATA	1-Wire	+3.3V (4.7 kΩ R61 Pull-up)
U18 / J9 (Moisture)	MOISTURE_SLEEP	Power Gated	TPS22919 via GPIO41
Q1/Q2 / J5/J6 (LEDs)	LED_PWM1 / PWM2	Low-Side PWM	+BATT Rail (R59/R60 68 Ω)

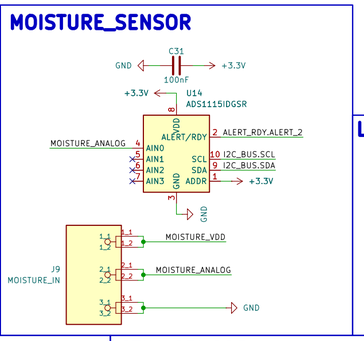
REFF-ELECTRODE



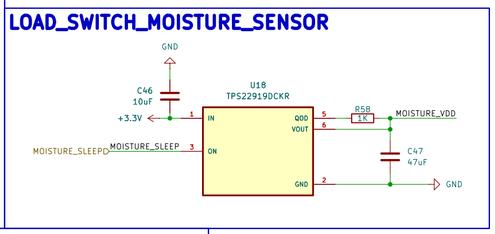
RTC



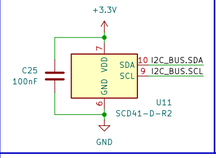
MOISTURE_SENSOR



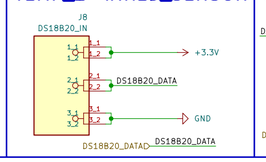
LOAD_SWITCH_MOISTURE_SHIELD



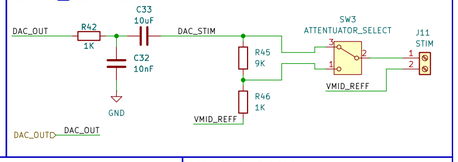
CO2_SENSOR



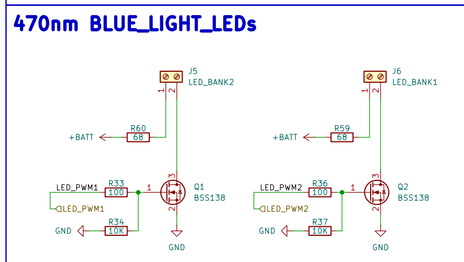
TEMP_1-WIRED_SENSOR



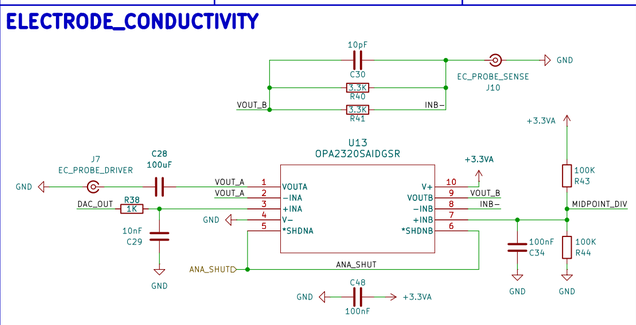
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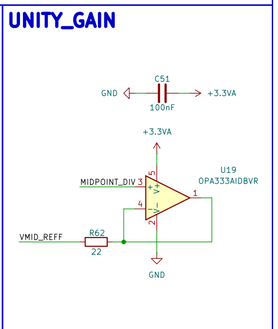
470nm BLUE_LIGHT_LEDs



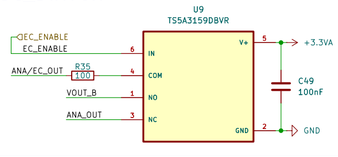
ELECTRODE_CONDUCTIVITY



UNITY_GAIN



ANALOG_SWITCH



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Size: A2	Date: 2026-10-25	**Rev: 3.6**
KiCad E.D.A. 10.0.7		Id: 8/10

LORA SUBSYSTEM DESIGN NOTES (LORA BLOCK)
Project: OpenMycoNet – COMBO_NODE
Status: READY FOR MANUFACTURING (JLCPCB 4-Layer Stackup)
Date: 2026-10-25
Revision: 3.6

SYSTEM OVERVIEW
The LoRa subsystem provides long-range Sub-GHz wireless communication using an SX1262 transceiver.
It features an integrated LC power-amplifier bias feed, a Johanson matched balun, a PE4259 RF switch for Tx/Rx path multiplexing,
and a coaxial antenna interface (J12). Communication and control are handled over the shared SPI bus and dedicated control signals
connected to the ESP32-S3.

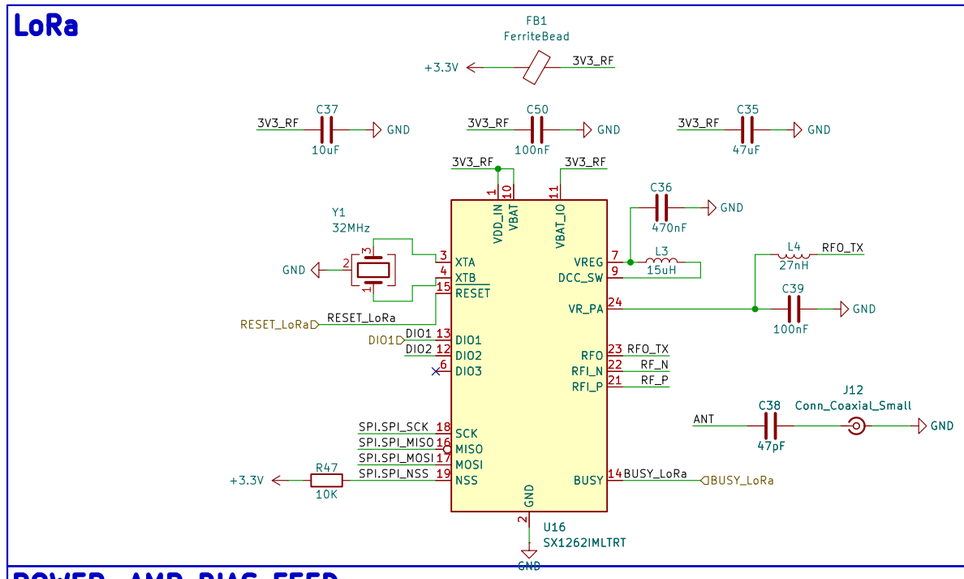
1. TRANSCEIVER CORE AND POWER MANAGEMENT (U16 SX1262IMLTRT)
* Core Transceiver: U16 (SX1262IMLTRT Sub-GHz LoRa/FSK transceiver).
* Primary Supply Filtering: Main +3.3V rail filetered by FB1 to produce clear 3V3_RF Rails decoupled
  by C50 (100nF), C33 (10µF) and bulk C37 (47µF).
* Core Voltage Rails: All the power pins, powered directly onto 3V3_RF rails (VDD_IN, VBAT, VBAT_IO)
  : VREG pin (pin 7) decoupled by C36 (470nF); VR_PA pin (pin 24) decoupled by C39 (100nF) to GND.
* DC-DC Converter: Internal switching converter DCC_SW pin (pin 9) connected to L3 (15µH power inductor) for high-efficiency power regulation.
* Reference Clock: Y1 (32MHz crystal unit) connected across XTA (pin 1) and XTB (pin 2).

2. RF FRONT-END, BALUN, AND ANTENNA PATH
* Power-Amp Bias Feed: RFO_TX pin (pin 23) powered via L4 (27nH RF choke inductor) connected to the VR_PA rail.
* Integrated Balun (FL2 0900FM15K0039): Converts differential RF_N (pin 22) and RF_P (pin 21) transceiver lines into
  a single-ended signal. Output passes through C44 (47pF) to generate the RF2_PATH (Rx) signal.
* Transmit Path: RFO_TX output routes through C40 (47pF) to generate the RF1_PATH (Tx) signal.
* RF Switch (U15 PE4259): Multiplexes Tx and Rx paths to the antenna. RF1_PATH connects to pin 1, RF2_PATH connects to pin 3,
  and CTRL (pin 4) is driven by the DIO2 line from the SX1262. Pin 5 (RFC) outputs the combined ANT signal.
* Antenna Interface: Single-ended ANT line routes through DC-blocking capacitor C38 (47pF) to coaxial connector J12.

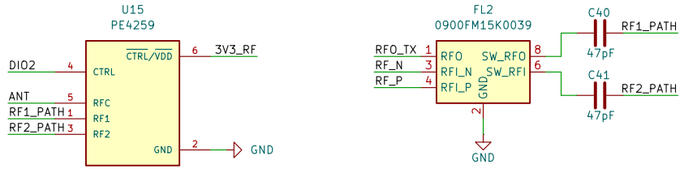
3. CONTROL LOGIC AND SPI BUS INTERFACE
* Shared SPI Bus: Communicates via SPI.SPI_SCK (pin 18), SPI.SPI_MISO (pin 16), and SPI.SPI_MOSI (pin 17).
* Chip Select Isolation: SPI.SPI_NSS pin (pin 19) pulled HIGH to +3.3V via R47 (10kΩ) resistor to ensure proper bus selection.
* Hardware Reset: Active-LOW RESET pin (pin 15) tied to RESET_LoRa net from ESP32-S3 (IO7).
* Busy & Interrupt Lines: BUSY pin (pin 14) outputs BUSY_LoRa status to ESP32-S3 (IO10). DIO1 pin (pin 13)
  outputs DIO1 interrupts to ESP32-S3 (IO11).
* Switch Control: DIO2 pin (pin 12) directly drives the PE4259 CTRL pin for automatic Tx/Rx path switching.

4. SUBSYSTEM INTERFACE MATRIX
SPI.SPI_NSS: Active-LOW chip select line (Pulled HIGH via R47).
SPI.SPI_SCK / MISO / MOSI: Shared hardware SPI bus lines connected to ESP32-S3.
RESET_LoRa: Transceiver hardware reset line connected to ESP32-S3 (IO7).
BUSY_LoRa: Transceiver busy status line connected to ESP32-S3 (IO10).
DIO1: Interrupt request signal line connected to ESP32-S3 (IO11).
J12: Coaxial antenna connector for external Sub-GHz antenna.

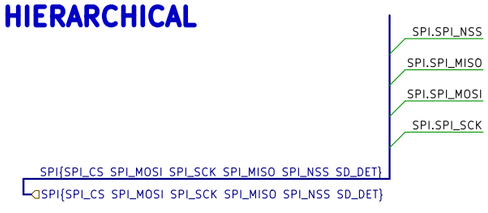
LoRa



POWER-AMP BIAS FEED



HIERARCHICAL



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File: LoRa.kicad_sch

Title: LoRa

Size: A3	Date: 2026-10-25	**Rev: 3.6**
KiCad E.D.A. 10.0.7		Id: 9/10

STORAGE MICROSD CARD SUBSYSTEM DESIGN NOTES (MICROSD BLOCK)
Project: OpenMycoNet COMBO_NODE
Status: READY FOR MANUFACTURING (JLCPCB 4-Layer Stackup)
Date: 2026-10-25
Revision: 3.6

SYSTEM OVERVIEW
The MicroSD card subsystem provides non-volatile data logging using a Hirose DM3AT push-push receptacle (J13) operating in 1-bit SPI bus mode. The interface features a high-side load switch (U17 TPS22919) for power gating, controlled output discharge timing, hardware card detection, power decoupling, and SPI bus pull-up resistors.

1. HIGH-SIDE POWER GATING LOAD SWITCH (U17 TPS22919DCKR)
* Core IC: U17 (TPS22919DCKR ultra-low leakage load switch with Smart Quick Output Discharge).
* Gated Supply Rail: Switches main +3.3V system input (decoupled by C44 10µF) to supply the SD card VDD rail (SD_VDD). Eliminates 100 µA – 1 mA idle card current during sleep.
* Control Line (SD_CARD_SLEEP): Driven by ESP32-S3 IO47 (shared with TPS3851 SET1 watchdog enable pin). Pulled LOW via 100kΩ resistor (R29) on MCU sheet to keep power disabled by default.
* Controlled Discharge Timing: External resistor R54 (1kΩ) connected between QOD (Pin 5) and SD_VDD (Pin 6), paired with C43 (47µF bulk capacitor), enforces a controlled fall time (t_FALL ≈ 152 ms) upon turn-off to prevent supply bus glitches.

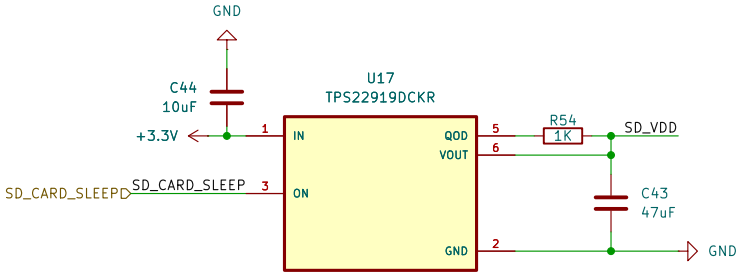
2. CONNECTOR FOOTPRINT AND POWER DECOUPLING (J13 HIROSE DM3AT)
* Memory Connector: J13 (Hirose DM3AT MicroSD receptacle with integrated mechanical card-detect switch).
* VDD Power Rail: Pin 4 (VDD) connected directly to the switched SD_VDD rail from U17.
* Local Decoupling: Filtered by C42 (100nF) and C43 (47µF) ceramic capacitors to handle peak card-write current spikes and maintain rail stability.
* Grounding: Pin 6 (VSS) and outer metallic shield frame (SH) tied directly to system GND.

3. SPI SIGNALING AND BUS PULL-UP NETWORK
* Chip Select (SPI.SPI_CS): Pin 2 (DAT3/CD) acts as active-LOW chip select from ESP32-S3 (IO15); pulled HIGH to SD_VDD via R51 (10kΩ) to prevent unselected floating states.
* Master-Out Slave-In (SPI.SPI_MOSI): Pin 3 (CMD) receives data from ESP32-S3 (IO13); pulled HIGH to SD_VDD via R51 (10kΩ).
* Serial Clock (SPI.SPI_SCK): Pin 5 (CLK) receives clock pulses from ESP32-S3 (IO14).
* Master-In Slave-Out (SPI.SPI_MISO): Pin 7 (DAT0) outputs data to ESP32-S3 (IO12); pulled HIGH to SD_VDD via R52 (10kΩ).
* Unused Data Lines: Pin 1 (DAT2) and Pin 8 (DAT1) pulled HIGH to SD_VDD via R48 (10kΩ) and R49 (10kΩ) per SD physical layer specs.
* Leakage Prevention: All SPI GPIO lines configured to High-Z / Input mode prior to pulling SD_CARD_SLEEP LOW to prevent back-feeding ESD protection diodes when SD_VDD is powered OFF.

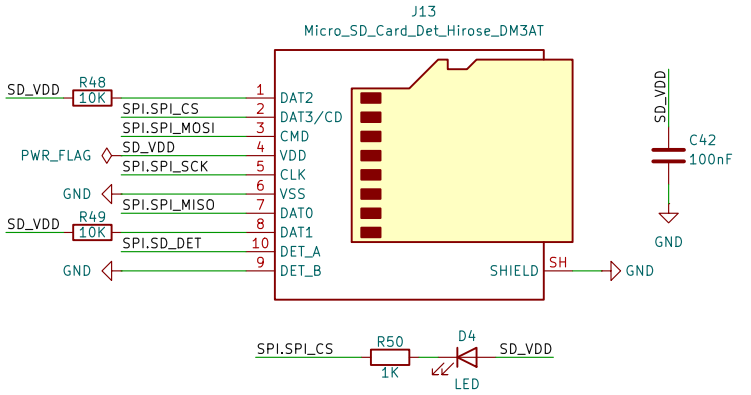
4. CARD DETECTION AND ACTIVITY INDICATOR
* Card Detect Circuit (SPI.SD_DET): Pin 10 (DET_A) pulled HIGH to SD_VDD via R53 (10kΩ); Pin 9 (DET_B) tied to GND. Inserting a card shorts DET_A to DET_B, pulling SPI.SD_DET LOW to signal insertion to ESP32-S3 (IO16).
* Activity LED: Active-LOW LED D4 connected via R50 (1kΩ) between SD_VDD and SPI.SPI_CS line; illuminates during active SPI read/write cycles.

5. SUBSYSTEM INTERFACE MATRIX
* SW_EN_SD: High-side load switch enable input from ESP32-S3 (IO47), pulled LOW via R55 (100kΩ).
* SPI.SPI_CS: Active-LOW Chip Select line connected to ESP32-S3 (IO15), pulled HIGH via R51 (10kΩ).
* SPI.SPI_MOSI: SPI data input line connected to ESP32-S3 (IO13), pulled HIGH via R51 (10kΩ).
* SPI.SPI_SCK: SPI clock input line driven by ESP32-S3 (IO14).
* SPI.SPI_MISO: SPI data output line connected to ESP32-S3 (IO12), pulled HIGH via R52 (10kΩ).
* SPI.SD_DET: Active-LOW mechanical card detect line connected to ESP32-S3 (IO16), pulled HIGH via R53 (10kΩ).

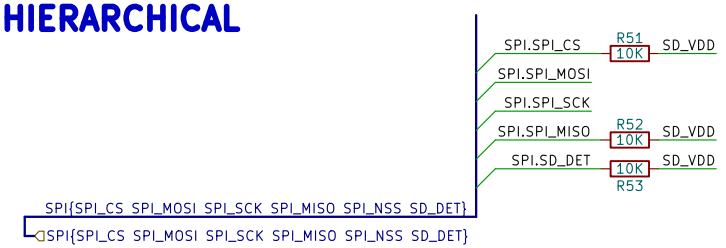
LOAD_SWITCH_SD



MEMORY_CARD



HIERARCHICAL



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File: Storage Micro SD Card.kicad_sch

Title: Storage Micro SD Card

Size: A4	Date: 2026-10-25	**Rev: 3.6**
KiCad E.D.A. 10.0.7		Id: 10/10